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A Dense Semantic Mapping System based on CRF-RNN Network

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***Abstract*— Geometric structure and appearance information of environments are main outputs of Visual Simultaneous Localization and Mapping (Visual SLAM) systems. They serve as the fundamental knowledge for robotic applications in unknown environments. Nowadays, more and more robotic applications require semantic information in visual maps to achieve better performance. However, most of the current Visual SLAM systems are not equipped with the semantic annotation capability. In order to address this problem, we develop a novel system to build 3-D Visual maps annotated with semantic information in this paper. We employ the CRF-RNN algorithm for semantic segmentation, and integrate the semantic algorithm with ORB-SLAM to achieve the semantic mapping. In order to get real-scale 3-D visual maps, we use the RGB-D data as the input of our system. We test our semantic mapping system with our self-generated RGB-D dataset. The experimental results demonstrate that our system is able to reliably annotate the semantic information in the resulting 3-D point-cloud maps.**

I. Introduction

Simultaneous Localization and Mapping (SLAM) is a technology which can simultaneously estimate robot poses and build modes for unknown environments. It has been studied for several decades, and acts as a fundamental capability for robots. In pioneer works, researchers used Lidar to do SLAM [1]–[3]. Nowadays, with the development of computer vision, Visual SLAM has become the mainstream. The task of Visual SLAM is to jointly estimate camera poses and construct environment models. Different from Lidar-based SLAM, Visual SLAM can provide more information. There are some well-known Visual SLAM systems which present highly impressive performance, such as [4]–[8].

Although Visual SLAM has come into existence for many years and gets more and more attention, many limitations are remaining to be solved. The work in the past were mostly focused on the pose estimation and mapping. With collected data from environments, robots can only build a map which contains only geometric and appearance information. It is sufficient for tasks such as map-based navigation and localization. However, for object-based navigation [9], such map cannot serve for robots very well. For example, to grab a cup, robots need to know not only *where the cup is?* but also *what is corresponding to a cup?*

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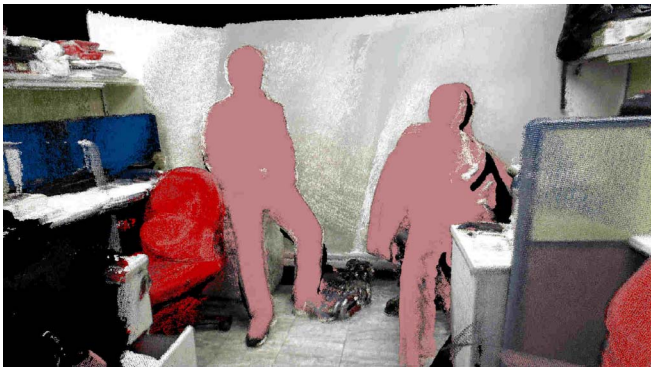

Fig. 1: An experimental result of our semantic mapping system. It shows a point-cloud map, in which the red points represent chairs, the blue points represent monitors and the pink points represent persons. For the objects we did not segment, we kept their original colors.

The ubiquitous and increasing GPU processing power makes deep learning an efficient tool in many robotic applications, such as Visual SLAM. The essential part of Visual SLAM is to get the observation from unknown environments and estimate camera poses. Traditionally, researchers conducted feature extraction and matching in Visual SLAM for pose estimation and mapping. With deep neural networks, we can go beyond this. The deep learning technology can be applied in Visual SLAM systems and we can understand the resulting maps with semantic information. Then, tasks such as object-based navigation become feasible.

In this paper, we propose a method that combines Visual SLAM and deep learning. Fig. 1 demonstrates an experimental result of our system. We use ORB-SLAM [8] as the SLAM system in our method. The Visual SLAM system can build an impressive 3-D map and estimate robot poses in unknown environments. Besides the tracking, local mapping and loop closing threads in the Visual SLAM system, we add one thread for semantic segmentation and one thread for point-cloud mapping. For the semantic segmentatin, we use the CRF-RNN [10] network. It takes as input 2-D images. With our mapping thread, we can get 3-D point-cloud maps. Note that like most semantic SLAM systems [11], the semantic information has not been used to enhance the SLAM system in our method. The major contribution of this paper is the integration of the ORB-SLAM system and the CRF-RNN network.

This paper is organized as follows. Section II describes existing Visual SLAM, semantic segmentation and semantic mapping systems. Section III presents the proposed mapping

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system. Section IV discusses the experimental results. The last section concludes this paper and discusses future works.

II. Related Work

In this section, we discuss several algorithms on Visual SLAM, semantic segmentation and semantic mapping. We focus on well-known or milestone works.

Visual SLAM can be divided into many branches. A remarkable early monocular SLAM system was the work presented by Davison *et al.* [12] . It was the first successful application of the SLAM methodology in the *pure vison* domain of a single uncontrolled camera. This system can recover the 3-D trajectory of a monocular camera. In 2014, Engel *et al.* [7] proposed LSD SLAM, which is a well-known monocular SLAM system. It allows to build large-scale, consistent maps of the environment. KinectFusion by Newcombe *et al.* [4] is a work for real-time dense surface mapping and tracking. Although it can conduct dense and accurate reconstruction, dense map is memory consuming and the system is only suitable for small-scale environments. In 2014, Endres *et al.* [5] presented a RGB-D SLAM system which requires a Microsoft Kinect. In 2016, Babu *et al.* [13] proposed a novel method called s-DVO. Different from sparse visual odometry, it makes fully use of all pixel information from an RGB-D camera. In contrast, Direct Sparse Odometry (DSO) omits the smoothness prior used in other direct methods and instead sampling pixels evenly throughout the images, with a trade-off, it can be real-time. These two systems can perform very well for large scale visual odometry, however, they cannot handle loop closure issues. ORiented Brief (ORB) is a feature descriptor and it is rotation invariant and resistant to noise. ORB-SLAM uses this feature and incorporates three threads that run in parallel. The three threads are tracking, local mapping, and loop closing. It demonstrates highly performance on the TUM RGB-D dataset.

There are many work about semantic segmentation. Many researchers address this problem with probabilistic graphical models. Researches on Markov Random Fields (MRFs) [14] and Conditional Random Fields (CRFs) [15] can be very representative. With the powerful tool Convolutional Neural Networks (CNN), many work are floured where using CNN to tackle with the semantic labeling problem. Works such as [16]–[18] have shown efficient performance for the semantic mapping. 3-D point-cloud semantic mapping, which means producing pixel-level labels, requires an accurate semantic segmentation tool, the word *accurate* means fine and smooth. Unfortunately, presence of max-pooling layers in CNN further reduces the chance of getting a fine segmentation output [19], which cannot guarantee the usefulness of the semantic map because tasks such as grabbing for robots require sharp boundary. To deal with this problem, Zheng *et al.* [10] proposed a new form of convolutional neural network that combines CNN with the CRF-based image segmentation algorithm.

There are many research works on semantic mapping. Hermans [20] proposed a novel 2-D to 3-D label transfer based on Bayesian updates and dense pairwise 3-D CRF, and the model presents speed advantage over other methods. The work by Salas-Moreno [21] which is known as *SLAM++* shows high accuracy object-level scene description. It takes advantage the prior knowledge that many scenes consist of repeated, domain-specific objects and structures. However, the objects it can label are limited to the ones in a predefined database. The work closely related to ours is the method proposed by zhao *et al.* [22]. It can build a dense 3-D map with a material label for each point. We argue that in some cases, we need object labels for the environment. In this paper, we focus on the integration of Visual SLAM and deep learning to achieve semantic mapping.

III. Method

As illustrated in Fig. 2, the pipeline of our method is composed of three modules. They are a feature-based SLAM system ORB-SLAM, a RNN network for semantic segmentation CRF-RNN, and a data association module. The role of ORB-SLAM is to recover camera trajectory and build 3-D point-cloud maps. It can provide the frame correspondence for the second module to decide which frame should act as the input of it. CRF-RNN receives a 2-D image and returns a pixel-wise labeling one. Finally, the data association, which registers the semantic information into the 3-D point-cloud map created by ORB-SLAM, acts as a integration tool. The following section presents the details of each modules.

A. Mapping

Mapping is a fundamental part of our framework. We choose ORB-SLAM, which is the state-of-the-art Visual SLAM system. It uses ORB [23] descriptor for feature extraction and matching, which is fast to compute and match and has shown good performance for place recognition. ORB-SLAM consists of three thread, namely, *Tracking*, *Local Mapping*, and *Loop Closing*. Given a new frame, tracking thread decides whether to insert it as a new key frame. Once a new key frame is inserted, the local mapping process new key frames and performs local BA to achieve an optimal reconstruction of the camera pose. The loop closing searches for loops with every new key frame. Once a loop is detected, a global graph optimization will be conducted. In addition, the system incorporates a bag of words place recognition module, based on DBoW2 [24], which helps to perform loop detection and relocalization. Besides the existing three threads, we add one thread for dense point-cloud mapping. Once a new key frame is decided, it will be passed to the added thread. This thread generates 3-D point-cloud maps.

B. Semantic Segmentation

The pixel labels of a given image can be modeled as a random field. In the fully connected pairwise CRF model, the energy of a label assignment x is given by:

$$E(x) = \sum_i \psi_u(x_i) + \sum_{i \neq j} \psi_p(x_i, x_j) \tag{1}$$

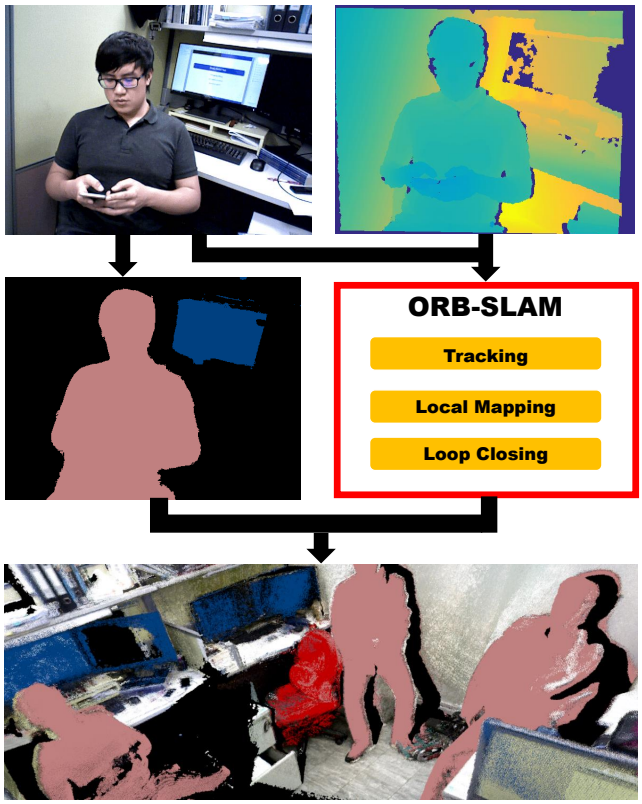


Fig. 2: The pipeline of our method. The inputs are RGB and depth images. RGB images are used for semantic segmentation. Depth data can help to build point-cloud maps. The segmentation results are registered to the point-cloud map.

where the unary energy component $\psi_u(x_i)$ measures the inverse likelihood of the pixel i taking the label x_i , and the pairwise energy component $\psi_p(x_i, x_j)$ measures the cost of assigning labels x_i, x_j according to pixels i, j. In CRF-RNN, unary energies are obtained from the CNN network.

Each iteration of the algorithm can be divided into five steps:

1) Using filters to conduct filtering each probabilistic graph for each class.
2) Summarizing the result of the filters with weights.
3) Transforming class compatibility.
4) Adding the unary potential.
5) Normalizing label probability for each pixel.

C. Data Association

The ORB-SLAM system computes a globally consistent trajectory. Our point-cloud mapping thread projects the original point measurements into a common coordinate frame. With the point-cloud map and the semantic segmentation result from 2-D RGB images, the role of data association is to register semantics to each point. Given a frame, the tracking thread determines whether it is a key frame. If it is a key frame, this frame will be inserted into SLAM and provided as the input of the CRF-RNN network. Then, the segmentation module outputs image with pixel-wise labels. If a key frame is determined, the system will use the semantic image to generate point cloud. We label each point with different colors. In case of regions in 2-D semantic image which do not have a label, we keep the original colors.

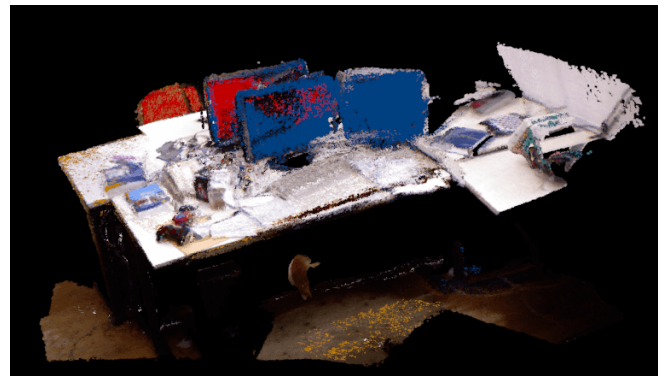


Fig. 3: The semantic mapping result of our system using the `fr1_desk` sequence. It shows a point-cloud map, in which the red points represent chairs, the blue points represent monitors and the pink points represent persons. For the objects we did not segment, we kept their original colors.

As indicated from [5], point-cloud maps cannot be updated efficiently in case of major corrections of the past trajectory as obtained by large loop closures. In most cases, it is reasonable to recreate the map in the case of such an event. ORB-SLAM is a system that allows little drift in a large scale scene, so the effect can be smaller.

IV. Experimental Results

We conduct experiments on three sequences of the TUM dataset and an office dataset recorded by ourselves. Our SLAM mapping thread presents a real-time performance. However, the CRF-RNN network cannot achieve real-time performance. Our system ran on a computer, which has an i7-6700K CPU, 32GB memories, a GTX 1070 GPU. The experimental results show that the median time of semantic segmentation process for each 2-D image is 1.5s.

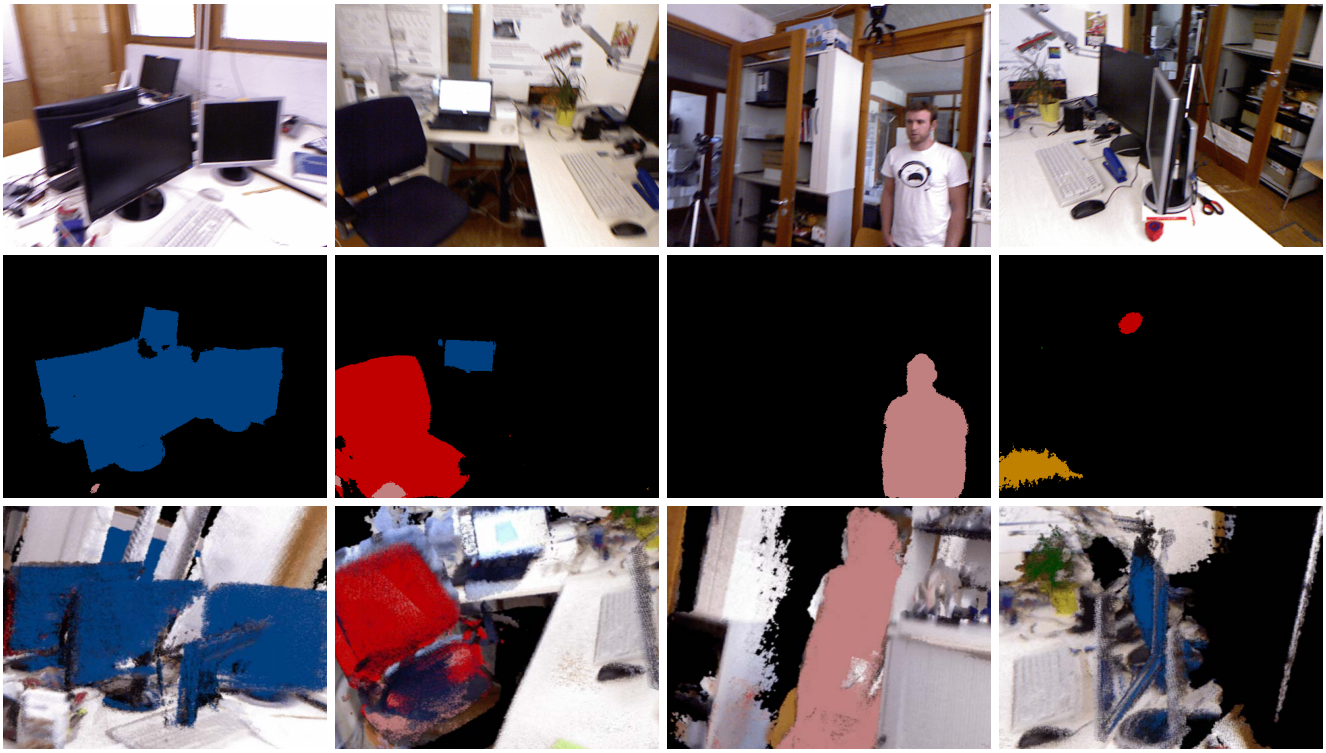
A. TUM Dataset

The TUM RGB-D dataset [25] contains associated RGB and depth images. The images are obtained from a Microsoft Kinect camera. The data was recorded at 30Hz and the 640×480 resolution. The first scene is an office. It starts with four desks and continues around the wall of the room until a loop is closed.

The second scene shows a desk in a laboratory. In this smaller scene, our system shows very good performance. The duration of this sequence is 23.40s, the average translational velocity is 0.413m/s, and the average angular velocity is 23.327deg/s with the trajectory dimension: 2.42m×1.34m×0.66m. Loop closure contributes a lot in the scene, and there is little drift.

B. Office Dataset

We produced a small experimental RGB-D dataset, which contains a loop that can be used to test the performance of



(a) (b) (c) (d)

Fig. 4: Sample experimental results using the `fr1_room` sequence of the TUM RGB-D dataset. The sub-figures from the top row to the bottom row are RGB images, semantic segmentation results and semantic mapping results. The red color represents chairs. The blue color represents monitors. The pink color represents persons. This figure is best viewed in color.

(a) (b) (c) (d)

Fig. 5: Sample experimental results using the `cuhk_office` sequence generated by ourselves. The sub-figures from the top row to the bottom row are RGB images, semantic segmentation results and semantic mapping results. The red color represents chairs. The blue color represents monitors. The pink color represents persons. This figure is best viewed in color.

TABLE I: The quantitative evaluation results of our method using the TUM and our sequences. The unit for the numbers in the table is %. The first row of the table shows the sequence names. The second row shows the object names.

	fr1_room			fr1_360			fr1_xyz			cuhk_office		
	Monitor	Chair	Person	Monitor	Chair	Person	Monitor	Chair	Person	Monitor	Chair	Person
FPR	1.31	5.24	0.50	4.83	7.41	1.12	0.00	10.34	0.50	0.00	10.29	0.00
FNR	6.31	10.22	1.72	8.18	14.53	3.21	10.20	30.50	44.32	14.22	30.53	5.51
Re	85.11	70.00	91.17	76.69	67.84	81.27	89.68	76.59	68.50	88.44	71.98	94.32
Pr	97.15	88.00	98.40	93.35	84.67	95.59	95.20	82.44	60.73	94.58	78.90	98.26
PWC	12.45	20.40	5.53	15.41	25.19	10.00	8.87	31.54	23.56	11.62	33.37	3.19

both SLAM and semantic mapping. The dataset has the same format with the TUM dataset. It is a complete indoor scene and there are chairs, persons, monitors which can be segmented very well with our semantic segmentation algorithm. Note that in our dataset, the persons kept still during the recording. We named our dataset as the cuhk_office dataset.

C. Qualitative Results

Fig. 4 and Fig. 5 qualitatively demonstrate the experimental results using the two sequences. As we can see, our approach produces impressive semantic mapping results. In semantic segmentation results and semantic mapping results, some objects are misclassified. This is caused by information insufficiency and ambiguity. In Fig. 4, some monitors only occupy a small region of the image. Insufficient information reduces the precision of semantic segmentation and semantic mapping. In Fig. 5, a person sitting in a chair leads to ambiguity whether there is a chair or not. This issue also reduces the precision of our results.

D. Quantitative Results

The section presents the quantitative Results. Note that we only take the objects: monitor, chair and person into account. We firstly count the following numbers. Let's take the monitor for example.

- TP (True Positive): The number of frames that there is monitor in the frame and the monitor has been correctly segmented.
- FP (False Positive): The number of frames that there is no monitor in the frame but some object has been segmented as monitor.
- TN (True Negative): The number of frames that there is no monitor in the frame and no object has been segmented as monitor.
- FN (False Negative): The number of frames that there is monitor in the frame but no object has been correctly segmented as monitor.

We use the following metrics for the quantitative evaluations. False Positive Rate (FPR), False Negative Rate (FNR), Precision (Pr), Recall (Re) and Percentage of Wrong Classifications (PWC). They are calculated as follows,

$$FPR = \frac{FP}{FP+TN} \tag{2}$$

$$FNR = \frac{FN}{TP+FN} \tag{3}$$

$$Re = \frac{TP}{TP+FN} \tag{4}$$

$$Pr = \frac{TP}{TP+FP} \tag{5}$$

$$PWC = \frac{FN+FP}{TP+FN+FP+TN} \tag{6}$$

For *Re* and *Pr*, the higher values are corresponding to better performance. For *FPR*, *FNR* and *PWC*, the higher values are corresponding to worse performance.

Tab. I presents the evaluation results using these metrics. As we can see, the values of *FPR* are small. This demonstrates that if there are some objects in the sequence, our method can precisely segment them out. The values of *FNR* are larger than those of *FPR*. This is because insufficient and ambiguous information degrades the performance of our approach. The values of *Re* and *Pr* are impressively high and those of *PWC* are low. This demonstrates that our model has high precision on the tested dataset. Note that our approach shows highly performance for persons, due to the large amount of of training data for them.

The CRF-RNN is an efficient network but it still has some limitations. Firstly, images in sequence may not be clear enough to produce a well segmentation result for the model mentioned above. Then, the model does not make use of the depth information. With impressive RGB-D dataset [26]–[30], the model should present better performance using both RGB and depth information. Finally, the efficiency must be improved for the real-time requirement.

V. Conclusions

We proposed here a CRF-RNN network-based semantic mapping system. The motivation of this paper is to equip Visual SLAM systems with semantic annotation capability. The proposed approach was divided into three modules. We employ the CRF-RNN algorithm for semantic mapping, and integrate the semantic algorithm with ORB-SLAM to achieve the semantic mapping. Qualitative and quantitative evaluations were carried out using the public TUM RGB-D dataset and our dataset. The results show that our system is able to reliably annotate the semantic information in the resulting 3-D point-cloud maps. However, our approach still presents some limitations. For instance, when object information is insufficient or ambiguous, the precision of our model will be degraded. When the camera moves fast through a scene,

our method cannot reliably reconstruct a semantic 3-D point-cloud map. To overcome these limitations, we would like to incorporate depth information to conduct more precise semantic segmentation. More training data would be used for highly precise semantic segmentation results. In addition, we would conduct the work that semantics could benefit the performance of SLAM systems.

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